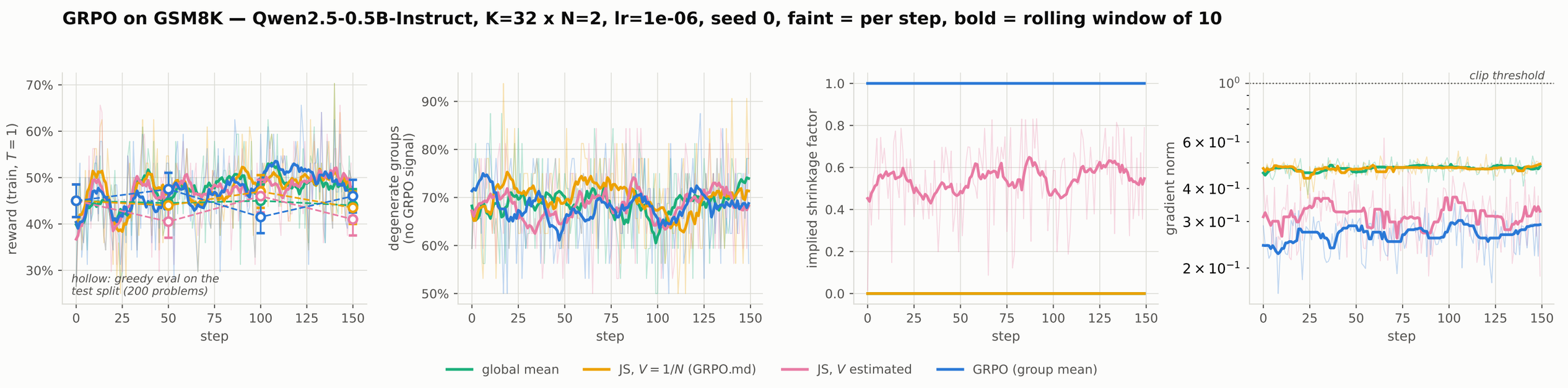

GRPO on GSM8K — Qwen2.5-0.5B-Instruct, K=32 x N=2, lr=1e-06, seed 0, faint = per step, bold = rolling window of 10
reward (train, T = 1)
70%
60%
50%
40%
30%
hollow: greedy eval on the
test split (200 problems)
0
25
50
75
100
125
150
step
degenerate groups
(no GRPO signal)
90%
80%
70%
60%
50%
step
implied shrinkage factor
1.0
0.8
0.6
0.4
0.2
0.0
step
gradient norm
clip threshold
10^{0}
6×10^{-1}
4×10^{-1}
3×10^{-1}
2×10^{-1}
step
global mean
JS, $V = 1/N$ (GRPO.md)
JS, V estimated
GRPO (group mean)